is rarely covered by insurance and can be a burden for patients undergoing curative treatment. In late 2022, with the rising price of gas, we set out to evaluate whether gas prices & rising costs from inflation increased financial toxicity for patients undergoing radiation treatment.

Materials/Methods: A prospective survey study was developed and approved by IRB as exempt study at two sites. The survey consisted of basic demographic question and functional assessment of chronic illness therapy (FACIT) Cost measure, which was licensed for the purpose of this study. Baseline information including distance to treatment center (as measured by zip code), average cost of gas in the state, income, education level, and transportation method were also collected. Distance from treatment, average gas price, average weekly gas cost, annual household income and insurance type were dichotomized. T test and fisher exact test were used to assess the differences between groups for ordinal and binary survey questions, respectively. A p-value <0.05 was considered significant.

Results: A total of 30 complete patient questionnaires were collected between August and November 2022. When gas prices rose over $4, the percent of patients who felt in control of their financial situation was significantly decreased (p = 0.049) and percentage of patients for whom gas prices have become a burden for radiation treatment significantly increased (63.2% vs 36.8%, p<0.001), and correlated to patient's feeling of financial stress, satisfaction with their current financial situation, and perception of choice on health care spending. Travel distance >10 miles to treatment center was correlated to feeling out of pocket medical expenses to be higher than expected and ability to meet monthly expenses. Household income >$40,000 significantly increased patient's ability to meet monthly expenses (p = 0.024) and feeling of control of personal financial situation (p = 0.004) and correlated to higher satisfaction with financial stability. Type of insurance did not significantly impact financial toxicity. Only 10% of the patient knew of financial assistance while 3% utilized financial support for general medical bill and/or travel support.

Conclusion: Rising gas prices impact patients' wellness. Rising prices are correlated to feeling more financial toxicity independent of income level. Further studies are needed to determine impact of inflation on financial toxicity. More emphasis is needed to identify high risk patients and offer more directed financial assistance.

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Trends in the Utilization and Outcomes of Brachytherapy vs. Proton Therapy in Localized Uveal Melanoma in the United States

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Purpose/Objective(s): To determine the trends in the utilization of different techniques of radiation treatment in localized uveal melanoma in the United States and to determine the difference in outcome between brachytherapy (BT) and proton-beam therapy (PT).

Materials/Methods: Using the North American Association of Central Cancer Registries (NAACCR), we identified 17451 patients with localized uveal melanoma with known both surgical and method of radiation therapy between 1995-2018. Univariate and multivariable analyses were performed to identify factors and characteristics that may have a potential impact on the treatment. Net survival was calculated using Relative survival in Ederer II cumulative expected method with actuarial. Overall survival (OS) and cancer-specific survival (CSS) were estimated by Kaplan-Meier method and life-tables.

Results: The registries included 28.312 patients with uveal melanoma, out of them 22,983 patients with localized disease. Patients with known both surgical and method of radiation therapy were 17451 patients. We have studied this group further. The median age of this group was 61 (IQR 51-71), males were 51.9% (N = 9063), whites were 98.1% (N = 16921), and choroidal tumors were 87.8% (N = 15330). The median tumor thickness was 3.9mm (IQR 24-70), while, the median maximum tumor diameter was 11.3mm (IQR 8.0-14.4). The median follow-up time was 5.9 years (IQR 3.1-10.3). Among all patients, 57.5% (n = 10032) received BT, and 9.8% (n = 1708) received PT. Collectively, 30.1% (n = 5247) of the patients received no radiation treatment. BT constituted 14% of treatment modalities in 1995 and steadily increased to 65% in 2017. PT utilization was minimal in 1995 (1.4%) and also increased to 14.3% in 2017. Younger age and thinner tumor thickness were independently associated with better Cancer-specific survival (CSS) and overall survival (OS). For all patients who were treated primarily with radiation treatment, the 5-year OS was 85,9% and the 10-year OS was 77.2%. There were no differences in rates of OS or CSS between patients who were treated with BT or PT.

Conclusion: There is a continuous increase in the utilization of both BT and PT in the treatment of localized uveal melanoma. Similar survival outcomes could be anticipated for both radiation treatment techniques.

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Plasma EBV DNA in Nasopharyngeal Cancer (NPC) Treated with Definitive Radiotherapy (RT)

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Purpose/Objective(s): EBV DNA has well-studied roles in NPC including early detection and surveillance. There are limited North American data on EBV DNA testing. Our center has used EBV DNA testing since 2010. We hypothesized: (1) higher first post-RT EBV DNA level is associated with worse prognosis, and (2) surveillance EBV DNA is specific for recurrence at a low detection threshold.

Materials/Methods: We retrospectively reviewed all patients with non-metastatic (TNM-7 stage I-IVB) NPC treated with definitive RT/chemoRT (CRT) ± adjuvant chemotherapy (AC) between 2010-2017. EBV DNA was assayed by quantitative PCR in a CAP/CLIA-certified laboratory and reported in copies/mL of plasma. Pre-RT is defined as 0-90 days before the first RT fraction and post-RT within one year after RT. We report log odds ratios (LOR) from a linear model of T- and N-category with log-adjusted EBV DNA as the response variable. Survival outcomes were analyzed with log-rank tests and Cox multivariate analyses (MVA) adjusted for age, stage, and treatment, reporting hazard ratios (HR). A total of 95% confidence intervals of LOR and HR are reported. The detection threshold that maximized the F1 accuracy score was considered optimal.

Results: Of 271 patients in the study window, 179 had pre-RT +/- post-RT EBV DNA testing. Six received RT, 43 CRT, and 130 CRT+AC. With 7-yr median follow-up, 37 recurred and 37 died. Detectable pre-RT EBV DNA was found in 154 (86%) with a median of 928 copies/mL (range: 1-239214). EBV DNA level correlated with higher N category (LOR: 0.28, 0.15-0.42, p<0.001), but not T category (0.04, -0.06-0.13, p = 0.5). Above-median pre-RT EBV DNA was associated with worse recurrence-free survival (RFS) by log-rank test (p = 0.016) and Cox MVA (HR: 2.2, 1.1-4.8, p = 0.03) along with N category, age, and no AC. Post-RT EBV DNA was available in 99 patients at a median of 54 days. RFS, progression-free survival (PFS), and overall survival (OS) were worse in patients with detectable post-RT EBV